

Figure 1.16 The D3 `select` syntax modifies style using the `.style()` function, and traditional HTML content using the `.html()` function.

1.3.2 Coding in the console

You'll do a lot of your coding in the IDE of your choice, but one of the great things about web development is that you can test JavaScript code changes by using your console. Later you'll focus on writing JavaScript, but for now, to demonstrate how the console works, copy the following code into your console and press Enter:

```
d3.select("div").style("background","lightblue").style("border", "solid black
     1px").html("Something else maybe");
```

You should see the effect shown in figure 1.16.

You'll see a few more uses of traditional HTML elements in this chapter, and then again in chapter 3, but then you won't see traditional DOM elements again until chapter 8, where we'll use D3 to create complex, data-driven spreadsheets and galleries using `<div>`, `<table>`, and `<select>` elements. If all D3 could do was select HTML elements and change their style and content like this, then it wouldn't be that useful for data visualization. To do more, we have to move away from traditional HTML and focus on a special type of element in the DOM: SVG.

1.3.3 SVG

A major value of HTML5 is the integrated support for SVG. SVG allows for simple mathematical representation of images that scale and are amenable to animation and interaction. Part of the attractiveness of D3 is that it provides an abstraction layer for drawing SVG. This is because SVG drawing can be a little confusing. SVG drawing instructions for complex shapes, known as `<path>` elements, are written a bit like the old LOGO programming language. You start at a point on a canvas and draw a line from that point to another. If you want it to curve, you can give the SVG drawing code coordinates on which to make that curve. So if you want to draw the line on the left, you'd create a `<path>` element in an `<svg>` canvas element in your web page, and you'd set the `d` attribute of that `<path>` element equal to the text on the right: